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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 03 – Lecture 03: Dimensionality Reduction (PCA, LDA, ICA)

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Topic focus: **reduce dimensions** for speed, noise reduction, and visualization

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

Quick Links

Why Reduce? PCA LDA/ICA Example Summary

Agenda

Why reduce dimensions?

PCA

LDA and ICA (overview)

Mini example

Summary

Learning Outcomes

By the end of this lecture, you should be able to:

- Explain why high-dimensional data can be difficult (curse of dimensionality).

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- Outline PCA steps and interpret explained variance.

Learning Outcomes

By the end of this lecture, you should be able to:

- Explain why high-dimensional data can be difficult (curse of dimensionality).
- Describe PCA intuition (maximum variance directions).
- Outline PCA steps and interpret explained variance.
- Distinguish PCA vs LDA vs ICA at a conceptual level.

Prerequisite Refresh

- From Lecture 01: features are design choices, and scaling changes the representation.

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- From Lecture 02: preprocessing must be fit on the **training split only**.

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- From Lecture 01: features are design choices, and scaling changes the representation.
- From Lecture 02: preprocessing must be fit on the **training split only**.
- PCA is a feature-transformation method, so both lessons still apply here.

Abbreviations & Symbols

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- PC: principal component, EVR: explained variance ratio

Curse of Dimensionality (Intuition)

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- Distances can become less informative.
- Models may overfit and require more data.

Dimensionality reduction helps with **compression**, **noise reduction**, and **visualization**.

PCA: Intuition

Principal Component Analysis (PCA)

Find orthogonal directions (components) that capture **maximum variance** in the data.

- First component: direction with highest variance.

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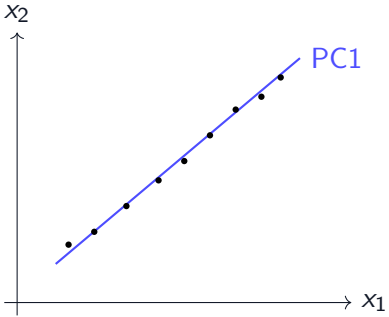
- First component: direction with highest variance.
- Next components: highest remaining variance, orthogonal to previous.

PCA Geometry (2D Intuition)



PCA chooses the direction with maximum spread (variance) and projects data onto it.

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5. Project data onto these components.

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- Centered data: $X_c = X - \mu$

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- Eigen-decomposition: $Cv_i = \lambda_i v_i$
- Projection (keep top-k): $Z = X_c V_k$

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Manual Procedure: Eigenvalues and Eigenvectors

1. Compute the covariance matrix C from centered data.

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4. Normalize each eigenvector.
5. Sort by decreasing eigenvalue: largest λ gives PC1.

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Worked 2D Example (By Hand)

Suppose:

$$C = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Then:

$$\det(C - \lambda I) = \det \begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix} = 0$$

$$(3-\lambda)^2 - 1 = 0 \Rightarrow (\lambda - 4)(\lambda - 2) = 0$$

So the eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 2$.

Worked 2D Example: Principal Directions

For $\lambda_1 = 4$, solving $(C - 4I)v = 0$ gives:

$$v_1 \propto \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \hat{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 2$, solving $(C - 2I)v = 0$ gives:

$$\hat{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Since $\lambda_1 > \lambda_2$, the diagonal direction $\hat{v}_1$ is the first principal component.

Explained Variance

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Explained Variance

- Eigenvalues indicate how much variance each component captures.
- Explained variance ratio helps choose k.
- Goal: keep enough components to retain most information.

Example rule: choose the smallest k with $\geq 90\%$ explained variance (depends on use-case).

Choosing k (Scree Plot Idea)

- Plot EVR per component: large drop $\Rightarrow$ "elbow" point.

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- Plot EVR per component: large drop $\Rightarrow$ "elbow" point.
- Or use cumulative EVR: pick smallest k reaching a target (e.g., 90%).
- Trade-off: compression vs information loss.

PCA vs LDA vs ICA (Concept)

- **PCA**: unsupervised; maximizes variance.

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- **PCA**: unsupervised; maximizes variance.
- **LDA**: supervised; finds projections that separate classes.
- **ICA**: unsupervised; finds statistically independent components (source separation idea).

Quick Comparison Table

Method	Uses labels?	Goal	Typical use
PCA	No	maximize variance	compression/visualization
LDA	Yes	separate classes	classification features
ICA	No	independent sources	signal separation

Treat LDA as a supervised preview here; its meaning becomes clearer after the classification lecture in Unit IV.

Mini Example (2D to 1D Projection)

- Suppose points lie roughly along a diagonal line in 2D.

In the lab, you will implement PCA and visualize explained variance.

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Mini Example (2D to 1D Projection)

- Suppose points lie roughly along a diagonal line in 2D.
- PCA finds the diagonal direction and projects data to 1D.
- Result: compressed representation with minimal information loss.

In the lab, you will implement PCA and visualize explained variance.

Leakage-Safe PCA in Python (1/2)

```python
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

X_train, X_test = train_test_split(X, test_size=0.2, random_state=42)

pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("pca", PCA(n_components=2))
])
```

First set up the train/test split and define a pipeline that standardizes before PCA.

Leakage-Safe PCA in Python (2/2)

```
Z_train = pipe.fit_transform(X_train)
Z_test = pipe.transform(X_test)
print(pipe.named_steps["pca"].explained_variance_ratio_)
```

Fit scaling and PCA on **training data only**; then reuse the same transformation on test data.

Summary

- Dimensionality reduction combats high-dimensional issues and improves efficiency.
- PCA finds maximum-variance orthogonal directions; explained variance guides k.
- LDA uses labels to separate classes; ICA seeks independent components.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

Exit Questions

1. What does PCA optimize?
2. Why do we often standardize before PCA?
3. Compare PCA and LDA in one paragraph.
4. What does "explained variance" mean?